Knowledge-Based Systems 333 (2026) 115027

Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys

Court to conversation: Tactical badminton analysis via computer vision and RAG-enhanced LLMs

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ARTICLE INFO

Keywords:
Sports analytics
Knowledge engineering
Computer vision
Shot classification
Conversational AI

ABSTRACT

This paper presents an original, innovative knowledge-based system designed to support human decision-making in the complex, dynamic domain of elite sports. Current tactical analysis in badminton often relies on subjective intuition. To address this, we introduce an end-to-end framework that transforms unstructured monocular video into a queryable knowledge base of tactical insights. The system's primary methodological contribution is a knowledge presentation and engineering pipeline that uses computer vision (YOLO, MediaPipe) to automatically extract and structure spatio-temporal events and player kinematic data. Building on this, we define and detect a novel key performance indicator, the tactically critical 'out-of-position' state, a measure of player vulnerability. This structured data forms a dynamic knowledge base, which is then reasoned over by a Retrieval-Augmented Generation (RAG) enhanced Large Language Model (LLM). This creates a novel intelligent decision support system, allowing expert users (coaches and athletes) to query the knowledge base with complex natural language questions and receive narrative, data-backed answers. The principal results demonstrate a robust, validated system that pioneers a new paradigm for interactive, AI-driven tactical discovery. Our major conclusion is that this knowledge-based approach provides a powerful tool for bridging advanced computation with practical, real-world expert analysis.

1. Introduction

A significant challenge in modern artificial intelligence lies in transforming vast amounts of unstructured video data into structured, queryable knowledge. This is particularly difficult in dynamic domains where complex, high-speed human actions are central. Elite sports offer a perfect testbed for this problem, as match videos contain a wealth of spatio-temporal data that, if properly structured, can be used to support human decision-making and tactical planning [1].

Badminton is an exemplar of such a domain, characterized by rapid exchanges and intricate movements that demand exceptional technical skill and tactical acumen [2]. Traditionally, performance analysis in this area has relied on manual video review, a foundational coaching practice [3] that is inherently subjective and unscalable.

This paper presents an original, innovative knowledge-based system designed to address this challenge. We introduce a comprehensive, end-to-end framework that transforms unstructured monocular video into a dynamic knowledge base of tactical insights. Answering a coach's complex query, such as "Why am I losing long rallies when pushed to my backhand?", requires more than simple event detection. It demands a system that can model and integrate player kinematics, court position, and the sequences of actions that lead to a state of tactical vulnerability [4].

Our framework achieves this through two main components. First, a computer vision pipeline performs knowledge extraction, using object detection (YOLO [5]) and pose estimation (MediaPipe [6]) to generate a structured dataset of spatio-temporal events. Second, this dataset serves as a knowledge base for a Retrieval-Augmented Generation (RAG) system [7], creating a novel intelligent decision support system. This allows expert users (coaches) to query the knowledge base with natural language and receive data-backed, narrative answers.

The primary contributions of this work are therefore:

- A novel **knowledge engineering pipeline** that uses fine-tuned computer vision models to automatically extract and structure tactical events and kinematic data from monocular sports video.
- A robust, multi-cue method for automated shot detection that combines trajectory and pose data, reducing reliance on standard action recognition models.
- The definition and automated identification of a novel key performance indicator: the tactically critical **'out-of-position' state**, a model of player vulnerability based on biomechanical markers.

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https://doi.org/10.1016/j.knosys.2025.115027
Received 2 September 2025; Received in revised form 10 November 2025; Accepted 29 November 2025
Available online 1 December 2025

- The implementation of a **conversational query interface** using a RAG-enhanced LLM, creating an intelligent decision support tool for expert users.

The remainder of this paper is organized as follows. Section 2 reviews relevant prior work. Section 3 details our proposed technical framework. The evaluation protocol and experimental setup are described in Section 4. Section 5 presents the quantitative and qualitative results of our analysis. Finally, Section 6 discusses the findings and limitations, and Section 7 concludes the paper and outlines future work.

2. Related work

Sports video analysis is an active and growing research area, significantly driven by advancements in computer vision and machine learning. Early approaches in many sports often relied on specialized, high-cost infrastructure, such as multi-camera frameworks [8] or Hawk-Eye style systems, which use multiple synchronized cameras to accurately track objects in 3D. While these systems provide highly accurate data, their high infrastructure and operational costs limit their accessibility to elite-level broadcasting and tournaments.

Monocular video analysis, which uses a single camera feed, offers a more practical and accessible alternative, but introduces the challenge of inferring 3D information from a 2D plane. In badminton, research in this area has focused on fundamental tasks such as player tracking and, most notably, shuttle tracking. Tracking the shuttle is particularly challenging due to its small size, extremely high speeds, non-parabolic flight trajectory, and frequent occlusion by players or the net [9,10]. While early methods used traditional computer vision techniques, recent approaches have increasingly adopted deep learning-based object detectors like YOLO [5] and SSD [11] to improve detection robustness. The development of advanced object detection methodologies, such as multi-step and zero-shot approaches, remains an active area of research in top-tier journals like Knowledge-Based Systems [12]. The development of robust, generalizable multi-object tracking systems for sports remains a key research challenge, with state-of-the-art methods focusing on overcoming issues like occlusion and motion blur [13].

Beyond simple tracking, automated event detection, specifically recognizing the strokes played, is essential for any tactical analysis. This has been approached from multiple angles. Some studies utilize wearable sensors on players' rackets or wrists to classify shots based on inertial data [14], but this requires instrumenting players. In the video domain, methods have leveraged deep learning for action recognition, using architectures like CNNs and LSTMs to classify shots such as smashes, drops, and clears directly from image sequences [15]. This aligns with a broader trend of applying machine learning to analyze tactical strategies in sports, such as analyzing spatio-temporal tactics in team sports [16]. Others have used skeleton-based action recognition, analyzing the movement of a player's body keypoints to identify a stroke [17]. However, these methods often face challenges. They may struggle with variations in player technique or camera angles, and critically, they often identify the type of stroke without its tactical context (e.g., distinguishing an attacking clear from a defensive one). Furthermore, some systems still require a manual verification step to correct errors, hindering full automation [18]. To address these challenges, our proposed 3-stage shot detection method moves away from traditional action recognition models. Instead, it combines cues from shuttle trajectory, player pose, and object proximity. This approach is conceptually aligned with state-of-the-art research in human-object interaction (HOI) tracking, where combining pose-aware features with object locations has proven effective for recognizing complex interactions in challenging, occluded video, including sports-specific actions like racket swings [19,20].

Identifying tactical states and patterns from positional data is a more complex task than simple event detection. A key enabling technology for this is markerless motion capture using human pose estimation. This field has seen rapid advancement, with state-of-the-art research focusing on enhancing the accuracy of kinematic analysis in dynamic sports environments [21,22]. Crucially, recent work has validated the reliability of markerless systems for measuring the joint kinematics used to identify states of imbalance and poor recovery during athletic movements [23].

While this technology provides a robust foundation for objective analysis, a research gap remains in applying it to engineer a holistic knowledge-based system for expert users. Commercial systems like Clutch AI [24] focus on basic automated tracking, while patents typically describe models for discrete tasks like action classification [25]. The automated detection of a nuanced, holistic tactical state like 'out-of-position' therefore remains a challenge. To address this, our work introduces a novel intelligent decision support system that synthesizes multiple methodological contributions: a robust multi-cue shot detection method, the definition of the 'out-of-position' state, and a conversational query interface using a Retrieval-Augmented Generation (RAG) system. The application of RAG for querying complex, domain-specific knowledge is a key research theme in this journal [7], and our work presents a novel application of this paradigm, moving beyond its use in entertainment, such as for automated commentary [26], to the domain of analytical tactical discovery. This synthesis of robust event detection, deep tactical state analysis, and intuitive interaction presents a novel application of this paradigm as an intelligent decision support tool.

3. Methodology

Our tactical badminton analysis framework, illustrated in Fig. 1, transforms raw monocular match footage into a queryable knowledge base. The pipeline consists of several interconnected stages, beginning with CV Processing for low-level feature extraction, followed by Parameter Extraction to derive high-level semantic attributes. This information populates a Structured Tactical Dataset, which is then reasoned over by a RAG-enhanced LLM to produce conversational insights.

3.1. CV processing

The initial stage processes the raw video feed to identify and track key elements within each frame. This pipeline is designed to be fully automated, converting pixel data into foundational spatio-temporal features.

3.1.1. Object detection

Real-time object detection is performed using fine-tuned YOLO (You Only Look Once) models [5]. To optimize performance and address the distinct visual characteristics of different targets, our pipeline employs two separate, specialized models. The first is a Court Homography Model, trained specifically to detect the four outer corners of the badminton court, which provides stable reference points for subsequent geometric normalization. This model is only run when the camera angle changes to enhance efficiency. The second is a Game Object Model, which is fine-tuned on a custom dataset of annotated badminton footage to detect dynamic elements, including the players, their rackets, and the shuttle. This specialized model is critical for accurately identifying the shuttle, which presents a significant detection challenge due to its small size, high velocity, and variable appearance.

3.1.2. Pose estimation

To perform a kinematic analysis of player movement, posture and balance are captured using MediaPipe Pose [6], which estimates 2D skeletal keypoints (e.g., shoulders, elbows, wrists, hips) for each player in every frame. This foundational pose data serves as the input for downstream feature engineering.

3.1.3. Shuttle tracking

To handle the shuttle's high speed and potential for brief occlusions, we need to track the shuttle across frames. Unlike a ball, the shuttle has

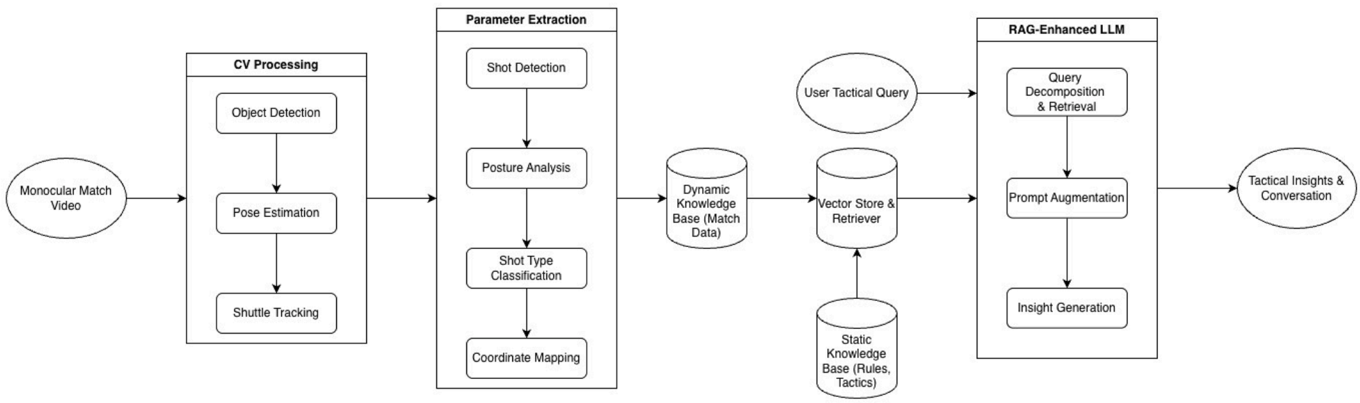

Fig. 1. The proposed system architecture for our knowledge-based framework. The pipeline transforms monocular video through a CV Processing stage and a Parameter Extraction stage to generate a **Dynamic Knowledge Base**. This is combined with a **Static Knowledge Base** of rules and tactics in a **Vector Store**, which is then queried by a **RAG-Enhanced LLM** to produce conversational tactical insights.

a non-parabolic trajectory and decelerates rapidly. Between successful YOLO detections, we use linear interpolation to estimate the shuttle's path, ensuring a continuous trajectory essential for subsequent analysis stages.

3.2. Parameter extraction

This stage uses the low-level outputs from the CV Processing stage to derive a set of high-level, semantic parameters for each event in the match.

3.2.1. Shot detection

A key methodological contribution is a robust, multi-stage method for automated shot detection. The process is a hierarchical cascade, initiated when a potential impact is inferred from the shuttle's trajectory. First, a *Trajectory Change Trigger* flags a potential shot event whenever the tracked shuttle's trajectory undergoes a significant change in direction or velocity. To validate this trigger and reduce false positives, a *Player-Shuttle Proximity and Pose Confirmation* step is performed. This step confirms the event by checking for close proximity between a player's dominant wrist, as identified by pose estimation (see Section 3.1.2), and the shuttle's location at the estimated time of impact. Finally, as a secondary verification, particularly in cases of ambiguous pose, we calculate the *Racket-Shuttle Intersection over Union (IoU)*. A high IoU between the bounding boxes of the player's racket and the shuttle provides strong, independent evidence of a shot.

3.2.2. Posture analysis

The skeletal keypoints extracted during Pose Estimation are further analyzed to derive key tactical features, namely shot handedness and the 'out-of-position' (OOP) state. Shot handedness is classified as either a forehand or a backhand by analyzing the relative positions of the body's centerline, shoulders, and the hitting wrist at the moment of impact. The primary focus of this analysis, however, is the modeling of the tactically crucial OOP state, which quantifies a player's tactical vulnerability. We define the OOP state as a boolean flag that is triggered if a player meets any of four distinct conditions at the moment of shuttle impact.

The first condition, a *Dive*, captures an emergency movement where the player's body is nearly horizontal, flagged if the torso's angle (from hip to shoulder midpoint) deviates more than $75°$ from the vertical axis. The second condition, *Off-balance*, identifies a significant loss of balance from a non-diving movement like a severe back-bend, and is triggered if the torso's deviation is between $40°$ and $75°$. The third condition, a *Stretch*, identifies a deep lunge that compromises recovery, flagged when the horizontal distance between a player's ankle keypoints exceeds 1.8 times their shoulder width. Finally, the fourth condition, *AWC (Away From Center)*, signals a recovery failure based purely on court positioning, triggered when the player's center of mass is more than 2 meters away from the court's central 'T' position at the moment when the opponent is playing the shot. These thresholds were determined empirically from observational analysis of our elite player dataset and form a multi-faceted, reproducible model for quantifying tactical disadvantage.

3.2.3. Shot type classification

Each confirmed shot is classified into a broader tactical category (e.g., Smash, Drop, Lift, Clear) by a rule-based classifier. This classifier synthesizes multiple spatio-temporal features to make its determination. The primary inputs include the player's court location at the moment of impact, such as their proximity to the baseline or net, which helps differentiate rear-court shots like smashes from forecourt shots. This is combined with the shuttle's height at impact relative to the net, a key indicator for distinguishing overhand from underhand strokes. Finally, the resulting shuttle trajectory immediately after impact provides crucial evidence; for example, a steep downward path is characteristic of a smash, whereas a high, arching trajectory indicates a clear or a lift.

3.2.4. Coordinate mapping

To enable a valid and objective analysis of player positioning and shot placement, a homography matrix, $\mathbf{H}$, is computed using the detected court corner coordinates. This matrix projects image pixel coordinates, $\mathbf{u} = [u, v, 1]^T$, to standardized, top-down real-world court coordinates, $\mathbf{x} = [X, Y, 1]^T$, as shown in Eq. (1).

$$\mathbf{x} = \mathbf{H}\mathbf{u} \tag{1}$$

This normalization is fundamental for analyzing shot placement and player positioning independent of camera perspective.

3.3. The hybrid knowledge base

The core of our system's memory is a hybrid knowledge base that synthesizes two distinct sources of information. This architecture is designed to provide both the specific factual data of a single match and the general domain context required for deep tactical reasoning.

The first component is the *Dynamic Knowledge Base*. This is the direct output of our CV and Parameter Extraction pipelines, resulting in a structured tactical dataset where each row represents a single shot event from the analyzed match. It contains all the extracted parameters such as player position, shot type, and the 'out-of-position' state. This dynamic base provides the specific, factual grounding for any user query. An excerpt is shown in Table 1.

The second component is the *Static Knowledge Base*, a separate, pre-curated collection of documents containing general, historical, and

Table 1
Excerpt from the extracted dataset showing key tactical and positional parameters for each shot in a rally. Note: P_ID refers to Player ID; Pos refers to the player's (X,Y) coordinates; Q refers to the Court Quadrant; OOP stands for Out-of-Position, describing states like a defensive stretch or being off-balance; AWC refers to the OOP state of being away from the center.

Rally	Shot	P_ID	Shot Type	Hand	P1 Pos	P1 Q	P2 Pos	P2 Q	Shuttle Pos	Outcome	Tactical State
2	1	2	Serve	Forehand	(3.5, 3.0)	6	(2.6, 13.0)	8	(2.6, 13.0)	During	-
2	2	1	Net Shot	Backhand	(4.0, 1.5)	3	(2.6, 11.0)	8	(4.0, 1.5)	During	In Position
2	3	2	Net Shot	Forehand	(2.6, 4.0)	5	(1.5, 12.5)	7	(1.5, 12.5)	During	In Position
2	4	1	Lift	Forehand	(4.0, 4.0)	6	(2.0, 10.0)	8	(4.0, 4.0)	During	In Position
2	5	2	Smash	Forehand	(2.6, 3.0)	5	(2.0, 6.0)	5	(2.0, 6.7)	During	P1-OOP (Stretch)
2	13	2	Block	Forehand	(3.8, 3.0)	6	(4.0, 8.0)	6	(4.0, 8.0)	During	P2-OOP (AWC)
2	14	1	Drive	Forehand	(3.0, 4.0)	5	(2.6, 8.0)	5	(3.0, 4.0)	During	In Position
2	15	2	Block	Backhand	(2.0, 5.0)	4	(1.0, 8.0)	4	(1.0, 8.0)	P1_Wins	P2-OOP (Off-balance)

expert domain knowledge about badminton. It includes the official rules, definitions of tactical concepts, and principles of strategy.

As shown in Fig. 1, both the structured data from the dynamic base (converted to text descriptions per shot) and the documents from the static base are chunked and indexed into a single, unified *Vector Store*. This unified store allows the retriever to perform a semantic search across both match-specific events and general tactical principles simultaneously.

3.4. RAG-enhanced LLM interface for decision support

The final stage of our framework is an intelligent decision support system powered by a Retrieval-Augmented Generation (RAG) interface [7]. This system's design addresses a key challenge: providing nuanced, context-aware answers that a simple database query cannot. To achieve this, the system reasons over a hybrid knowledge base that synthesizes two distinct sources of information: the dynamic knowledge base, which is the structured tactical dataset generated in real-time by our CV pipeline for the specific match being analyzed, and the static knowledge base, a curated collection of documents containing general, historical, and expert domain knowledge about badminton. It includes the official rules, court dimensions, definitions of tactical concepts, and information on common strategies and historical precedents. The dynamic base provides the specific, factual grounding for any query, while the static base provides the necessary context and rules for the LLM to interpret that factual data correctly. Both knowledge bases are chunked and indexed into a vector store using a sentence-transformer model. The interface operates through a multi-step data flow.

3.4.1. Query decomposition and retrieval

The process begins when a user submits a natural language question (e.g., "Why do I lose long rallies on my backhand?"). This critical first step translates the user's natural language query into an effective retrieval strategy. This is achieved not through fixed parsing rules, but by leveraging the reasoning capabilities of the Large Language Model itself in a "pre-query" step. The user's initial question is embedded into a prompt that instructs the LLM to act as a 'tactical analyst' and rephrase the query into a rich, descriptive statement optimized for semantic search. For instance, the example query might be transformed into a detailed statement like: 'Analysis of rally sequences longer than eight shots where Player 1 loses the point, focusing on the events immediately following a backhand stroke played by Player 1.' This transformed statement is then embedded and used to perform a vector search against the knowledge base, retrieving the most relevant data chunks for the next stage.

3.4.2. Prompt augmentation

Following retrieval, the system performs prompt augmentation. A detailed, final prompt is constructed for the LLM. This is the "Augmented" part of RAG. The prompt includes not only the user's original question but also the specific, contextual data chunks retrieved from the vector store in the previous step.

3.4.3. Insight generation

Finally, this rich, contextualized prompt is passed to the LLM for insight generation. The LLM processes the prompt, synthesizes the patterns within the provided data, and uses its reasoning capabilities to formulate a coherent, human-readable answer. This complete RAG approach significantly mitigates the risk of LLM "hallucination" and ensures that the generated insights are both relevant and directly grounded in the factual data from the analyzed match.

4. Experimental methods

4.1. Materials and implementation details

4.1.1. Dataset for evaluation

The framework was evaluated on a dataset comprising 5 full men's singles matches, totaling approximately 4.5 hours of footage and over 750 rallies. The matches were sourced from publicly available, high-definition (1080p) broadcasts on the official BWF YouTube channel [27]. To ensure a robust and diverse test of the system's capabilities, the matches were purposefully selected to ensure a variety of internationally recognized tactical archetypes were represented. The player pool therefore included athletes known for distinct playing styles, such as the powerful attacking of Viktor Axelsen and Shi Yuqi; the persistent retrieving and defensive play of Kento Momota; the creative and unorthodox shot-making of Anders Antonsen; and the fast-paced, aggressive flair of Anthony Ginting and Srikanth Kidambi. This sampling strategy ensured the dataset contained a rich variety of rally lengths and complex tactical situations, forming a challenging and realistic testbed for our framework. The videos also presented a realistic set of real-world challenges for our computer vision pipeline, including significant variations in camera angle, intermittent motion blur, and frequent player-shuttle occlusions.

4.1.2. Sample size and justification

This study is an exploratory investigation into a novel methodological framework. As the primary goal was to establish a proof-of-concept and validate the feasibility of the end-to-end pipeline, a formal a priori sample size calculation was not performed. The sample of 5 full matches was deemed sufficient to demonstrate the system's ability to operate across a wide range of real-world scenarios.

4.1.3. Implementation details

The core computer vision pipeline was implemented in Python, leveraging OpenCV, PyTorch, and MediaPipe. To enhance detection accuracy, we utilized a YOLOv8 model for game objects (players, rackets, shuttle) fine-tuned on a combination of publicly available datasets from Roboflow Universe [28,29]. The homography calculation was performed using another fine-tuned YOLOv8 model trained on a separate public court detection dataset [30]. The intelligent decision support system was implemented using the LangChain framework. The hybrid knowledge base was created by chunking the relevant texts and then indexing them into a FAISS vector store. For the embedding process,

Fig. 2. An example frame processed by the computer vision pipeline. The image illustrates the simultaneous detection of key elements required for tactical analysis: (1) **Court Keypoints (Green Circles):** Used to compute the homography for a top-down view. (2) **Player Pose (Purple Lines/Dots):** Skeletal keypoints are extracted for posture analysis and stroke classification. (3) **Shuttle (Green Box):** Detected and tracked to identify shot events and trajectories. This frame captures the system's ability to extract multiple concurrent data streams from a dynamic, in-game situation.

we utilized the 'all-MiniLM-L6-v2' sentence-transformer model from the Hugging Face library, a common and efficient choice for semantic retrieval tasks. For full reproducibility, all experiments were run using a fixed random seed of 42, with key library versions including PyTorch 2.9.0, OpenCV 4.11.0, and LangChain 0.3.20.

The conversational query interface leverages LangChain's built-in "RetrievalQA" chain, which interfaces with a Gemini Pro-model [31]. All model training and inference were performed on a single NVIDIA P100 GPU within a Kaggle notebook environment, using a batch size of 16 for object detection.

4.2. Evaluation protocol

The evaluation focused on two key areas: a quantitative validation of the core computer vision components and a qualitative assessment of the tactical insights generated by the intelligent decision support system.

4.2.1. Quantitative validation of pipeline components

To establish a ground truth for validation, 20 complete rallies were randomly selected from the dataset and manually annotated. This annotated subset was used to measure the performance of our system against human-verified data using several standard metrics. We evaluated Object Detection using Mean Average Precision (mAP), Shot Detection with Precision, Recall, and F1-Score, Court Mapping via the mean reprojection error of known court points, and Pose Estimation using the Percentage of Correct Keypoints (PCKh).

4.2.2. Parameter sensitivity analysis

To address the system's robustness and its sensitivity to key heuristic parameters, we conducted parameter analyses for both the shot detection module and the "out-of-position" (OOP) state definitions.

For the 3-stage shot detection module, the process relies on three key parameters. The initial step is the Trajectory Change Trigger, which flags a potential event based on a minimum change in the shuttle's direction (in degrees). This is followed by the primary confirmation step, the Player-Shuttle Proximity check, which validates the shot based on a maximum distance (in pixels) between the player's wrist and the shuttle. Finally, a Racket-Shuttle IoU threshold is used as a secondary verification in ambiguous cases. Each of these parameters governs a trade-off between precision (minimizing false positives) and recall (minimizing missed shots). The optimal value for each was determined through an iterative tuning process. Starting with a set of reasonable defaults, we optimized each parameter sequentially, using the newly found optimal value in the subsequent optimizations for the other parameters. This cycle was repeated until the entire set of parameter values stabilized. The sensitivity analysis in Table 2 was then conducted to validate this final, stable configuration. For each test shown, one parameter was varied while the others were held constant at their determined optimal values.

Table 2
Consolidated sensitivity analysis for key shot detection parameters. The selected value for each parameter used in our framework is highlighted in bold, representing the peak F1-Score from its individual test.

Parameter	Value Tested	Precision	Recall	F1-Score
Trajectory Change	10 degrees	0.84	0.99	0.91
(Min. Direction Change)	**20 degrees**	**0.93**	**0.88**	**0.90**
	30 degrees	0.97	0.79	0.87
Player-Shuttle Proximity	80 pixels	0.87	0.94	0.90
(Max. Distance)	**60 pixels**	**0.94**	**0.88**	**0.91**
	40 pixels	0.97	0.81	0.88
Racket-Shuttle IoU	0.30	0.88	0.91	0.89
(Min. Overlap)	**0.40**	**0.95**	**0.86**	**0.90**
	0.50	0.98	0.80	0.88

The analysis in Table 2 demonstrates that a clear optimal value exists for each parameter, with individual F1-scores peaking around 0.90-0.91. It is the synergistic combination of these individually optimized parameters within the complete 3-stage pipeline that enables our final system to achieve the robust F1-Score of 0.92 reported in Section 5.1. For each parameter tested, a lower or more permissive threshold maintains high recall but at the cost of significantly lower precision, reducing the overall F1-Score. Conversely, a higher or more restrictive threshold improves precision but causes a sharp drop in recall, effectively increasing the number of missed shots, which also lowers the F1-Score. This comprehensive validation demonstrates that our parameter selection is the result of a deliberate optimization process. To further test for negative interactions, we performed a joint perturbation check by varying the

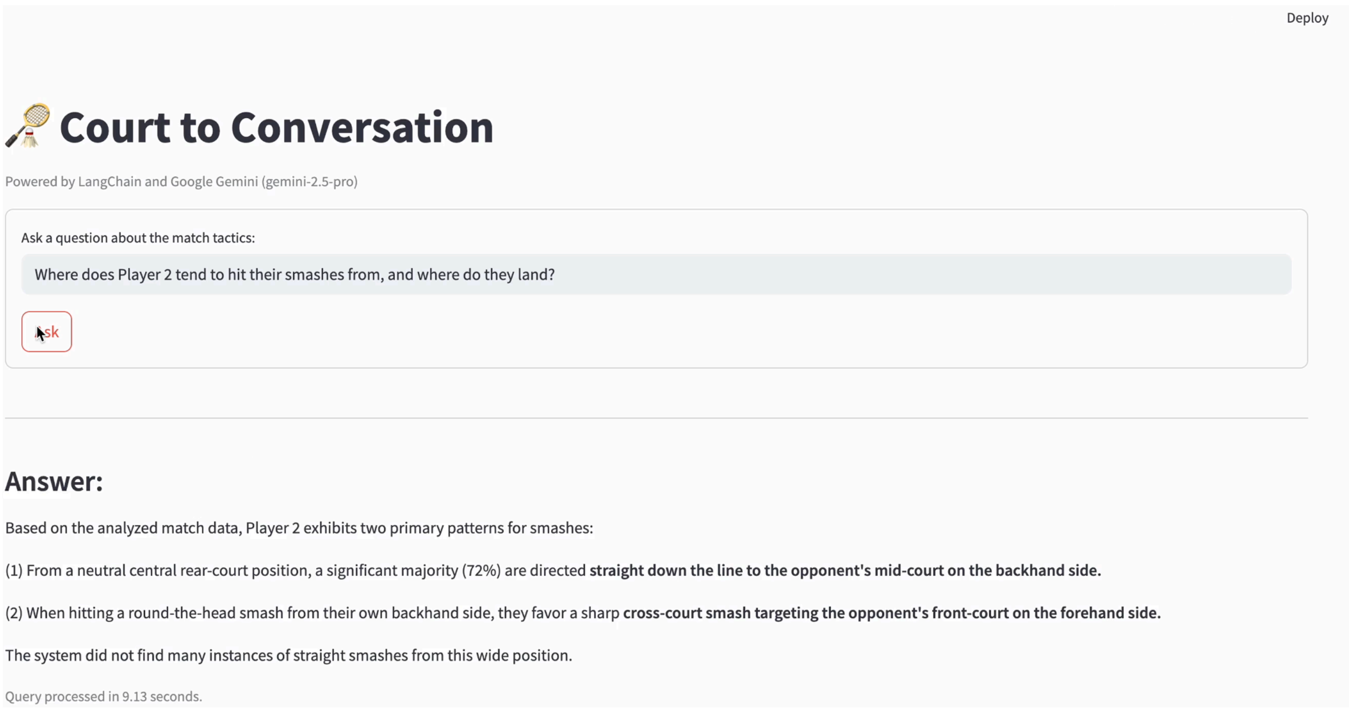

Fig. 3. Demonstrating the 'Court to Conversation' pipeline's final output, built using the Streamlit library in Python as a proof-of-concept. This figure shows a user's natural language question about tactical smash patterns and the corresponding data-driven insight generated by the RAG-LLM system, illustrating the framework's end-to-end capability.

two most sensitive parameters (Trajectory Change and Player-Shuttle Proximity) one step from their optimal values simultaneously; the resulting F1-score of 0.88 confirmed that system performance degrades gracefully and no unexpected interactions were observed.

For the OOP state definitions (e.g., Dive, Off-balance), the parameters are semantic thresholds based on player kinematics. Unlike the binary classification of a shot, a quantitative validation of these states would require a large-scale, multi-expert annotation effort, which is outlined as a key area for future work. Therefore, these thresholds were determined through a qualitative process combining observational analysis of our elite player dataset with established biomechanical principles of athletic movement. For instance, the 40–75 ° range for the "Off-balance" state was chosen after a qualitative sensitivity check. A stricter threshold (e.g., >50 degrees) failed to capture critical moments of instability during deep lunges, while a more permissive threshold (e.g., <35 degrees) incorrectly flagged routine athletic movements, introducing noise into the knowledge base. The practical validity of these chosen thresholds is further supported by the results of our domain expert evaluation (see Section 5.2), where tactical insights derived directly from these OOP states were judged to be highly plausible and relevant.

4.2.3. *Tactical insight generation evaluation*

The primary evaluation of the overall system's efficacy as a decision support tool involved assessing the quality and relevance of the insights generated by the RAG-LLM interface. This was performed qualitatively by posing various tactical queries to the system, representative of questions a coach or player might ask. Sample queries included "Describe the typical rally structure when Player A wins" and "In rallies lasting more than 8 shots, what typically happens right before I lose the point?". The responses were evaluated based on their relevance, factual accuracy (groundedness in the data), coherence, and insightfulness.

4.2.4. *Domain expert evaluation of tactical insights*

To assess the practical usability and tactical relevance of the generated insights beyond the authors" own evaluation, we conducted a formal qualitative assessment. The study involved a panel of 13 domain experts. The panel comprised a highly credentialed group of three certified, high-performance badminton coaches and ten elite-level competitive players, including a 2008 Olympian, a former Junior World No. 1, and several current and former top-100 world-ranked athletes.

Each expert was presented with the three sample query outputs detailed in the Results section (Section 5.3). They were then asked to evaluate the system's generated insights using a questionnaire with questions rated on a 5-point Likert scale. The evaluation focused on three primary criteria: *Tactical Plausibility* (how well the insights align with elite-level play), *Tactical Relevance* (how meaningful the insights are for performance), and overall *Practical Utility* (how useful the tool would be in a real-world coaching scenario). The full text of the sample insights and the complete questionnaire are provided in Appendix A.

5. Results

5.1. *Quantitative validation results*

The automated pipeline was successfully deployed on the evaluation dataset. The quantitative performance of the core computer vision components, evaluated against the manually annotated ground truth of 20 complete rallies, is summarized in Table 3.

The results indicate a high degree of accuracy and robustness across the pipeline. The object detection models achieved a high mAP of 0.94 for players and a robust 0.83 for the shuttle, demonstrating strong performance despite the challenges of motion blur and small object size. The novel 3-stage shot detection method proved highly effective, achieving an F1-Score of 0.92. This reflects a strong balance between a high

Table 3
Quantitative performance metrics for core pipeline components.

Component	Metric	Result
Object Detection		
	mAP@0.5 (Player)	0.94 ± 0.02
	mAP@0.5 (Shuttle)	0.83 ± 0.04
Shot Detection		
	Precision	0.95 ± 0.03
	Recall	0.90 ± 0.04
	F1-Score	0.92 ± 0.03
Court Mapping	Reprojection Error	3.4 ± 0.7 pixels
Pose Estimation		
	PCKh@0.5 (Upper Body)	0.93 ± 0.02
	PCKh@0.5 (Lower Body)	0.86 ± 0.05

Table 4
Average scores from the domain expert evaluation (n = 13) on a 5-point Likert scale.

Evaluation Criterion	**Average Score (out of 5)**
Tactical Plausibility	4.2
Tactical Relevance	4.3
Practical Utility	4.2

precision (0.95), which minimizes false positives, and a solid recall (0.90).

The system's geometric accuracy was validated by a low mean reprojection error of just 3.4 pixels, providing the necessary precision for reliable spatio-temporal analysis. Furthermore, the pose estimation performed well, particularly for the upper body (0.93 PCKh), providing a robust kinematic foundation for downstream analysis. The slightly lower score for the lower body (0.86 PCKh) reflects the well-known challenges of limb occlusion during the deep lunges and rapid footwork inherent to badminton.

5.2. Quantitative validation of the "out-of-position" state

A primary contribution of this work is the novel "out-of-position" (OOP) key performance indicator. To quantitatively validate its effectiveness as a measure of player vulnerability, we performed a correlation analysis on our entire dataset of over 750 rallies. We compared the outcome of a rally when an OOP state was detected versus when no OOP state was detected.

The analysis revealed a strong correlation between a player being in an OOP state and their probability of losing the rally. A chi-squared test for independence confirmed this association ($\chi^2(1, \mathrm{N}=1504) = 112.5$, $p < 0.001$). Overall, players had a baseline rally loss rate of 49.2 %. However, when a player was flagged with an "Off-balance" or "Stretch" state during a rally, their probability of losing that same rally increased to **76.8 %** (95 % CI [71.9 %, 81.7 %]). The odds ratio was 3.4, indicating that the odds of a player losing the rally were over three times higher when they were in an OOP state.

This result provides strong quantitative evidence that our heuristic-based OOP model is not arbitrary, but is a powerful and accurate predictor of tactical disadvantage and negative rally outcomes. It successfully quantifies the concept of player vulnerability, validating its use as a key performance indicator.

5.3. Domain expert validation of qualitative insights

The formal qualitative assessment described in Section 4.2.4 yielded highly positive feedback from our panel of 13 domain experts, confirming the system's potential as a valuable coaching and analysis tool. The quantitative ratings are summarized in Table 4.

As shown in the table, the experts found the system's observations to be highly plausible and tactically meaningful. Most significantly, when asked about the overall Practical Utility, 92 % of the experts rated the tool as a 4 (Useful) or 5 (Very Useful). The qualitative feedback provided rich context for these scores. Experts praised the system's ability to provide "a starting point for building tactics/strategy" and its value for "self game study." One expert summarized the system's core benefit, stating, "It's all about 'playing the percentages' and this could help doing that and in turn lead to being more efficient." The constructive feedback, such as the desire for linking insights directly to video clips and performing multi-match analysis, has been incorporated into our discussion of future work. Overall, the expert evaluation provides strong evidence for the real-world applicability and effectiveness of our "Court to Conversation" framework.

5.4. Qualitative results: a case study of the intelligent decision support system

The qualitative evaluation of the RAG-LLM interface demonstrated its efficacy as an intelligent decision support tool. The system was able to synthesize the extracted spatio-temporal data into coherent, data-backed answers to complex tactical queries. Below are two representative examples.

For the first example, the system was tasked with identifying the cause of failure in long rallies, with the following query:

> *"In rallies lasting more than 8 shots, what typically happens right before Player 1 loses the point?"*

The system generated the following insight, correctly identifying the 'out-of-position' state as the root cause:

> "Analysis of long rallies lost by Player 1 shows a recurring pattern. Often, two or three shots before the point ends, Player 1 is forced to play a defensive backhand lift from deep in their backhand corner. The data indicates these lifts frequently land in the opponent's mid-court, allowing the opponent to execute a steep attacking shot (smash or drop). The 'out-of-position' flag was triggered for Player 1 on 65 % of these defensive lifts, indicating poor recovery posture, which led to an inability to counter the subsequent attack."

For the second example, the system was asked to analyze a player's smash patterns:

> *"Where does Player 2 tend to hit their smashes from, and where do they land?"*

The system successfully synthesized positional data to identify two primary patterns:

> "Based on the analyzed match data, Player 2 exhibits two primary patterns for smashes: (1) From a neutral central rear-court position, a significant majority (72 %) are directed straight down the line to the opponent's mid-court on the backhand side. (2) When hitting a round-the-head smash from their own backhand side, they favor a sharp cross-court smash targeting the opponent's front-court on the forehand side. The system did not find many instances of straight smashes from this wide position."

These examples illustrate the system's ability to synthesize information across multiple data points (player position, shot type, shot destination, rally sequence, and tactical state flags) to identify actionable tactical patterns. The insights generated were evaluated to be relevant, factually grounded, and potentially valuable for strategic planning. This complete 'Court to Conversation' process is visualized in the following

Table 5
Qualitative comparison of "Court to Conversation" with other tactical analysis approaches.

System / Approach	Data Input Method	Primary Analytical Output	Computational Approach	User Interface
Hudl [32] (Traditional Tagging)	Labor-intensive manual tagging of video events	Video clips tagged with simple events (e.g., "smash")	N/A (Manual)	Video playlist or simple dashboard
Action Recognition (Deep Learning Models)	Monocular video	Probabilistic shot classification (e.g., "85 % clear") Critically lacks tactical context	Heavy (GPU-intensive, data-hungry)	N/A (Methodology only)
Clutch AI [24] (Emerging Commercial AI)	Monocular video	Basic event detection and player tracking (e.g., shot speeds)	Proprietary (often deep learning-based)	Visual analytics dashboard
TacticsFlow [33] (Visual Analytics Platforms)	Manually coded or pre-processed data strings	Heatmaps and visual representations of data patterns	Proprietary	Interactive dashboard
Court to Conversation (Our Framework)	**Fully automated from monocular video**	**Deep, contextual insights (e.g., 'out-of-position' state, causal patterns)**	**Lightweight (Heuristic-based, optimized for speed)**	**Conversational AI (Natural Language Queries)**

figures: Fig. 2 shows an example frame from the underlying computer vision analysis, while Fig. 3 demonstrates the final conversational output in the proof-of-concept interface. Video demonstrations of both components are available for review (see Section 8).

5.5. Conceptual ablation study: the impact of the 'out-of-position' feature

To validate the specific contribution of our novel 'out-of-position' (OOP) feature, we conducted a conceptual ablation study. We posed a complex, open-ended tactical query to the system and generated two responses: one using the full knowledge base, and one simulating a system where the OOP feature was removed. This comparison highlights the critical role the OOP state plays in moving beyond simple event description to provide deep, causal insights.

The following query, a classic analytical challenge for coaches, was posed to the system:

> *"In rallies lasting more than 8 shots, what typically happens right before Player 1 loses the point?"*

System Insight (with 'Out-of-Position' feature):

> Analysis of long rallies lost by Player 1 reveals two primary patterns of failure, both linked to poor recovery after being forced into a difficult position. The most common scenario occurs when the opponent forces Player 1 deep into their backhand corner, where they are often hitting a defensive lift while **severely off-balance or stretched**. This compromised position leads to a weak, short lift that sets up an easy, rally-ending smash. Additionally, a second pattern emerges after Player 1 plays a net shot. The opponent responds with a deep lift, forcing Player 1 to scramble backwards, and the data shows they are frequently caught **off-balance during this rapid retreat**. This unstable movement makes their subsequent defensive shot weak, allowing the opponent to win the point with their next attack.

System Insight (with 'Out-of-Position' feature removed):

> In rallies lasting more than 8 shots that Player 1 loses, an analysis of the shots played reveals two frequent scenarios.
>
> The most frequent scenario is straightforward: Player 1 plays a lift from the front-court, and the opponent wins the rally on the next shot with a smash.
>
> A second frequent scenario involves rallies where Player 1 plays a net shot. In these same rallies, the data shows that the opponent's winning shot is often a smash that lands in the deep court. The rallies where this occurs also frequently contain a lift played by the opponent.

This comparison starkly illustrates the impact of the OOP feature. The first response provides a rich, actionable diagnosis by identifying the root cause in both patterns: the player's compromised biomechanical state. It delivers a causal explanation that is invaluable for coaching. The second response, lacking the OOP context, can only provide a shallow, descriptive summary. It correctly identifies the shots present in the rally but is incapable of forming the crucial causal link between them. It presents a disconnected list of events, not an insight, demonstrating that the OOP feature is fundamental to the system's ability to perform meaningful, explanatory reasoning.

6. Discussion

This study demonstrates the successful development and validation of a novel, end-to-end knowledge-based system for the automated analysis of a complex, real-world domain. Our framework successfully bridges the gap between low-level computer vision analysis and high-level, human-centered decision support by generating a structured knowledge base from unstructured video and enabling natural language queries. The results validate our approach as a significant step towards creating more accessible and interpretable AI systems for expert domains.

6.1. Significance of methodological contributions

The key methodological contributions of this work have broader implications for the field of applied AI. The novel 3-stage shot detection method, which fuses trajectory, pose, and proximity cues, provides a robust heuristic-based alternative to data-hungry deep learning models for event detection in noisy, real-world video. This multi-faceted approach is a valuable technique for navigating challenging visual conditions.

Furthermore, the definition and detection of the 'out-of-position' state represents a key contribution to knowledge engineering. We have successfully modeled a complex, nuanced, and previously subjective human concept ('tactical disadvantage') using quantifiable kinematic and spatio-temporal features. This process of translating expert intuition into a machine-readable key performance indicator is a core challenge in the development of expert systems.

Finally, the RAG-LLM interface demonstrates a powerful paradigm for human-AI interaction. Its significance can be understood through a conceptual ablation. A simplified version of our system would replace the RAG-LLM with a direct, keyword-based search over the knowledge base. For a query like "Why do I lose long rallies on my backhand?", such a system would merely retrieve a list of all relevant shots, leaving the user to manually sift through raw data to find a pattern. In contrast, our RAG interface acts as an automated 'tactical analyst', capable of synthesizing patterns, performing implicit causal reasoning, and generating a coherent, narrative answer. This leap from raw data retrieval to automated insight generation is a core contribution of our framework.

Moreover, this RAG-based architecture provides an inherent mechanism for mitigating bias and ensuring factual grounding. Unlike a traditionally fine-tuned model whose responses can be influenced by opaque biases, our system's LLM is constrained to reason only over the information provided in its context. This context is drawn from our hybrid knowledge base: a dynamic base of factual match data and a static base of general badminton rules. This design choice ensures that every insight is auditable and directly traceable to a specific event in the match video or a rule in the static base, rather than being an ungrounded speculation.

6.2. Comparison with state-of-the-art approaches

To contextualize our framework, we provide a comparative analysis against existing approaches for tactical analysis in sports. As most commercial systems are closed-source, a direct quantitative benchmark is challenging. Therefore, Table 5 presents a qualitative comparison along several key methodological and feature-based dimensions.

As illustrated in the table, our “Court to Conversation” framework presents a novel synthesis of capabilities not found in other systems. It is unique in combining a fully automated pipeline from accessible monocular video with the ability to identify a deep, contextual tactical state (our ‘out-of-position’ indicator), rather than just simple events. Furthermore, our intentionally lightweight, heuristic-based design prioritizes speed and accessibility, making it suitable for rapid, mid-tournament analysis. Finally, the conversational RAG-LLM interface represents a significant paradigm shift from the traditional dashboards used by other systems, lowering the barrier to entry and allowing for a more flexible and intuitive exploration of tactical patterns.

6.3. Limitations and future directions

While promising, our system has limitations that span its entire architecture, from the foundational computer vision pipeline to the higher-level intelligent reasoning system. These challenges, combined with valuable feedback from domain experts, provide a clear roadmap for future work.

First, the system’s performance is fundamentally dependent on the upstream computer vision components, which inherit challenges inherent to monocular video analysis. Occlusions can still pose difficulties for tracking, especially for the shuttle as it passes behind a player. The accuracy of pose estimation can degrade under poor lighting or extreme motion blur, which in turn can affect the reliability of the ‘out-of-position’ (OOP) detection. Addressing occlusion in pose estimation is a significant and ongoing research challenge [34]. A key methodological limitation is that our OOP model is currently heuristic-based. A crucial next step is to transition to a fully data-driven model by training a classifier on a large, multi-expert-annotated corpus, which would not only create a more robust system but also represents an interesting research problem in learning complex, human-centric concepts from limited labeled data.

Second, while our RAG-based framework is designed to mitigate LLM hallucinations, the intelligent system has its own potential failure modes. The quality of a generated insight depends on the interaction between a query and the knowledge base, leading to several challenges: ambiguous user queries can yield unfocused answers; data sparsity for rare events can lead to over-generalization or an inability to identify patterns; contradictory evidence (e.g., a tactic succeeding and failing) may result in non-committal summaries; and semantic retrieval errors can cause the retriever to fetch irrelevant data, leading to a factually grounded but incorrect answer.

Beyond these operational challenges, the system’s aim to answer “why” questions positions it within the broader context of causal reasoning. The static knowledge base of rules acts as a “causal prior,” which presents a vulnerability: errors or oversimplifications within this prior can lead the LLM to misinterpret factual match data. For example, a rigidly defined tactic may cause the system to fail to recognize a creative deviation. This problem is conceptually aligned with mitigating prior errors in causal structure learning, a research area focused on building models resilient to flawed initial assumptions [35].

These limitations have been instrumental in shaping our future roadmap. Guided by the constructive feedback from our domain expert evaluation, we will focus on enriching the user experience by integrating automated data visualizations (e.g., heatmaps) and linking narrative insights directly to corresponding video clips. Another critical direction is to extend the knowledge base to perform multi-match analysis, identifying a player’s long-term tactical patterns. Technologically, our agenda includes the transition to a data-driven OOP model and drawing from causal reasoning research to develop methods for validating our static knowledge base and allowing the system to flag inconsistencies where observed data contradicts established priors. This user-centered development, combined with deeper technical and theoretical advancements, forms the core of our future research.

6.4. Broader impact and applications

This system has the potential to significantly augment traditional badminton coaching and player self-analysis. By providing objective, data-driven answers to specific tactical questions, it can help athletes and coaches identify patterns, strengths, and weaknesses that might not be apparent through observation alone. The efficacy of the tool in connecting analytical results with coaching practice is demonstrated by the proof-of-concept user interface (Fig. 3). This conversational format allows coaches to query the data intuitively, receiving immediate, narrative feedback to inform strategic decisions. For instance, a coach could use the system during a post-match review to instantly generate a data-backed summary of rally patterns, rather than relying solely on manual review. The structured dataset also provides a valuable resource for researchers interested in sports science, biomechanics, and data analytics.

While this system was developed and validated in the domain of elite badminton, its architectural principles have significant potential for broader impact. The core concept of using a CV pipeline to create a structured knowledge base from video, and then using a RAG-LLM to make that knowledge base queryable, is highly generalizable. This framework could be adapted to other sports (e.g., tennis, volleyball) or even to other expert domains that rely on the manual review of video, such as surgical training analysis, manufacturing process monitoring, or behavioral research. The conversational format lowers the barrier to entry for complex data analysis, providing a valuable model for future intelligent decision support systems across various fields.

7. Conclusion

In this paper, we presented “Court to Conversation”, a novel, end-to-end knowledge-based system for the automated tactical analysis of complex, real-world video data. Our framework successfully demonstrates how a synthesis of computer vision and natural language processing can transform unstructured video into a queryable knowledge base, creating a powerful intelligent decision support system for expert users.

The core of our work integrates several key methodological contributions: a robust, multi-cue shot detection method that reduces reliance on dedicated action recognition models; a novel system for modeling and identifying a tactically significant ‘out-of-position’ state from player kinematics; and a conversational query interface powered by a Retrieval-Augmented Generation (RAG) system. The successful validation of this pipeline represents a significant step towards creating more accessible and interpretable AI systems for domains that have traditionally relied on manual human analysis. By automatically generating detailed event data and enabling intuitive, natural language queries, our work opens new avenues for data-driven discovery and decision-making.

Building on the foundation of this framework, our future research roadmap is clear. As detailed in our Discussion, future work will proceed along three main avenues: transitioning from the current heuristic-based rules to a more robust, data-driven model; enhancing the system’s practical utility with user-centric features identified during our expert evaluation; and deepening the system’s reasoning capabilities by drawing from the field of causal structure learning.

8. Supplemental material

To provide a dynamic demonstration of the framework's capabilities, two supplemental videos are available on Zenodo. Video 1 demonstrates the computer vision pipeline processing a sample rally, and Video 2 demonstrates the conversational UI. The material is permanently archived and publicly available at the following DOI: https://doi.org/10.5281/zenodo.16940656.
Note: If a video does not load on the first attempt, please try refreshing the page once or twice.

CRediT authorship contribution statement

Kriti Bharadwaj: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization, Writing – review & editing; **Gowri Srinivasa:** Writing – review & editing, Supervision, Project administration, Conceptualization, Methodology.

Data availability

The datasets used in this study are publicly available on Roboflow and have been appropriately cited in the manuscript.

Data availability

The video footage used for analysis in this study was sourced from publicly available broadcasts on the official BWF (Badminton World Federation) YouTube channel [27]. The annotated image datasets used for training the computer vision models are publicly available on Roboflow Universe [28–30]. The final structured tactical dataset generated during the current study is available upon reasonable request. Supplemental videos demonstrating the system have been provided anonymously for review (see Section 8).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors acknowledge the use of Google's Gemini Pro-model (Version 2.5) for assistance with language editing, grammar enhancement, and refining the clarity of the manuscript. We are deeply grateful to all the domain experts who provided invaluable feedback for our user study. We extend special thanks to: Olympian and National Coach of the Singapore Badminton team, Anup Sridhar; Sagar Chopda, Head Coach at one of India's national academies; former Junior World No. 1 Varun Kapur; top-50 world-ranked player Tanya Hemanth; and elite players Purva Barve, Orijit Chaliha, and Annanya Pravin for their time and insights.

Appendix A. Domain expert evaluation materials

A.1. Sample insights presented to the expert panel

The following three system-generated insights were presented to the 13 domain experts for their evaluation.

Query 1: "where does player 2 tend to hit their smashes from, and where do they land?"
Based on the analyzed match data, Player 2 exhibits two primary patterns for smashes: (1) From a neutral central rear-court position, a significant majority (72 %) are directed straight down the line to the opponent's mid-court on the backhand side. (2) When hitting a round-the-head smash from their own backhand side, they favor a sharp cross-court smash targeting the opponent's front-court on the forehand side. The system did not find many instances of straight smashes from this wide position.

Query 2: "when is player 1 most often out-of-position?"
Analysis of the match data indicates that Player 1 is most frequently flagged as 'out-of-position' under specific tactical pressure. The most common scenario occurs when they are forced to play a defensive backhand lift from the deep, wide corner of the court. This situation typically arises after the opponent plays a cross-court drop shot, which pulls Player 1 forward and wide. The subsequent difficult recovery often results in a defensive posture, leaving them vulnerable to the next shot.

Query 3: "what is the most common rally pattern when Player 2 wins a point?"
The most common winning pattern for Player 2 involves a sequence of three shots. The pattern begins with Player 2 playing a deep clear to Player1′s backhand. This is followed by a short, weak return from Player 1. Player 2 then capitalizes on this with a winning smash to the open mid-court. This pattern was successful in 68.

A.2. Evaluation questionnaire

The following questions were presented for each of the three sample insights. All questions were rated on a 5-point Likert scale.

For each sample insight:
1. **Tactical Plausibility:** "Based on the example provided, how well do the system's observations align with what you would expect to see in real, elite-level badminton?" (1 = Highly Implausible, 5 = Highly Plausible).
2. **Tactical Relevance:** "How meaningful and important are these types of insights for a coach or player trying to understand tactical patterns and improve performance?" (1 = Not Relevant, 5 = Very Relevant).

Overall assessment:
3. **Practical Utility:** "Imagine a tool that could provide these kinds of automated insights. How useful would it be in your personal coaching or training workflow?" (1 = Not Useful, 5 = Very Useful).
4. An optional open-ended text box was provided for additional comments and quotes.

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